Ethan Lam

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Education

University of California, San Diego La Jolla, CA
B.S. Data Science, GPA: 3.63 *Expected: Jun. 2027*

Relevant Coursework: Machine Learning Algorithms, Probabilistic Models, Systems for Scalable Analytics, Data Management, Data Visualization, Recommender Systems & Webdata Mining, Statistical Methods

Experience

Data Science Intern Jun. 2025 – Sept. 2025
Pacific Life - Life Insurance *Newport Beach, CA*

- Automated competitor annuity rate retrieval using a Python data ingestion pipeline, reducing a previously **8-hour manual task** to **10 seconds**. (a 99.9% reduction in processing time)
- Designed and managed Snowflake tables using complex **SQL** queries, enabling reliable data preparation.
- Developed a **Gradient Boosting Regressor** and visualized predictions using **Tableau** to simulate annuity sales under shifting interest rate and stock market environments, enabling a data-driven competitive edge.
- Benchmarked 8 baseline models via **scikit-learn pipelines**, incorporating **outlier handling** (IQR capping, z-score filtering) and **variance reduction techniques** (feature scaling, PCA) to ensure robust generalization.
- Engineered advanced feature transformations (polynomial, market conditions) that drove a **27%** increase in predictive accuracy.

Projects

AI-Powered Earnings Call Preparation Dashboard | *Streamlit, Financial Analysis, Claude API* Feb. 2026

- Developed an AI-backed analytics app to analyze multi-source financial data, generating predictive Q&A that reduced earnings call preparation time by **94%**.
- Processed 100+ PDF documents to extract 200,000+ words of unstructured financial text using PDFIntelligence.
- Built predictive model analyzing YoY trends (NRR, Revenue, FCF) against competitor benchmarks to forecast analyst concerns.
- Developed an interactive business intelligence dashboard with Streamlit featuring a Question Generator, Financial Performance Dashboard, Document Search, and PDF Library, enabling quick and simple queries.

Severe Power Outages EDA & Prediction Model | *Model Development, Evaluation, Reinforcement* Dec. 2024
GitHub | Website

- Conducted a comprehensive data analysis of a public data set consisting of severe power outages in the United States using statistical analysis and data visualizations to identify data patterns and relationships.
- Assessed the missingness mechanisms of several features and performed permutation tests to verify dependency.
- Developed machine learning pipelines using scikit-learn to implement a **Random Forest** model with optimal hyperparameters using **GridSearchCV** for a total of 9 features based on climate, electricity usage, and more.
- Tested model accuracy/precision/recall by using K-Fold cross validation, yielding a **70% precision rate**.
- Concluded that Severe Power Outages are the result of poor maintenance

N-Gram Language Model | *Natural Language Processing, Web Scraping* Nov. 2024

- Scraped a corpus of literary text by web scraping a public eLibrary using RegEx and BeautifulSoup.
- Built a Uniform and Unigram Language Model, leveraging the corpus to generate sentences.
- Designed and implemented a N-Gram Language Model incorporating conditional probabilities of word sequences to generate coherent and meaningful text.
- Trained model on The Complete Works of Shakespeare, successfully and efficiently generating a 100-word passage in under 30 seconds.

Technical Skills

Technologies: Python, SQL, R, Spark, Hadoop
Developer Tools: Git, AWS, Snowflake, Azure Dev Ops, Tableau, Power BI, Streamlit
Libraries: pandas, NumPy, scikit-learn, Snowpark, Matplotlib, Plotly, Seaborn, D3.js, PyTorch
Other Skills: Statistical Inference/Analysis, Data Wrangling, Feature Engineering, Model Evaluation, A/B Testing, Data Analysis, Data Validation, Data Modeling, Dig Data Query, Artificial Intelligence, Excel, PowerPoint